sentinel-oo: system manual

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11 November 2026

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sentinel-oo: system manual

What the system is, how it is put together, and why it is put together that way. Written for somebody who is going to change it, review it, or decide whether to adopt it.

If you only want to use it, read the user manual instead.

The diagrams are drawn images, so they appear in every viewer and in the PDF. They are generated by `diagrams/make_diagrams.py`, so a diagram that goes out of date is a script to edit rather than a picture to redraw.

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1. What it decides

A **handover** is the moment one phase of work passes its output to the next. In a Design, Build, Test, Learn cycle there are four of them, and this is a service that sits at one.

It answers exactly one question: **can the next phase act on what the previous phase produced?**

A crate goes in and a crate comes out. The crate that comes out is the one that went in, plus a record of what was checked, against which rules, by which build of this service, with which model if any, and what was found.

A **crate** here means an RO-Crate: a folder of data files with a JSON file alongside them describing what each file is and how the things in it relate to one another.

The three verdicts

Verdict	What it means	What should happen next
`ACCEPTED`	Nothing was found	The next phase proceeds
`ACCEPTED_WITH_FLAGS`	Nothing stops the handover, but something was recorded. A value a model recovered always lands here	The next phase proceeds and something or somebody reads the flags
`REFUSED`	At least one requirement with severity `refuse` was violated	The next phase must not proceed

What it is not

It is not a validator. A third party tool, `rocrate-validator`, already checks that the packaging is well formed, and this service does not duplicate that.

The distinction those two make is the whole point of the system. A crate can satisfy every rule a standard has and still be useless to the phase receiving it. Measured on 11 November 2026 with `rocrate-validator` 0.13.1, against the six crates of the worked example, with no model running:

Crate	`rocrate-validator`	`sentinel-oo`
The published example	valid, 0 issues	ACCEPTED
Profile declaration removed	valid, 0 issues	REFUSED
A gene calling setting removed	valid, 0 issues	REFUSED
A reference to a file it never describes	valid, 0 issues	REFUSED
The setting present, but only as a flag in a command line	valid, 0 issues	REFUSED
The setting present, written out in a sentence	valid, 0 issues	REFUSED

The last row is the one worth understanding. Nothing is missing from that crate. The value is there, inside the text of a command line, where no rule can read it. So a valid crate can be refused for being unreadable rather than for being incomplete.

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2. Three separate things

Keeping these apart is the central design decision, not a tidiness preference.

	What it is	Contains code?	Specific to a use case?
The product	The service and the deployment program	Yes, all of it	No
A deployment	Contract files and example data	No	Yes
The demonstration	A stand-in for the next real tool	Yes	Neither, it stands in for reality

Why it matters. Somebody from another Design, Build, Test, Learn cycle must be able to take the product on its own, write contract files for their own handover, and deploy, without writing or editing any code. That is the difference between a product and one team's project.

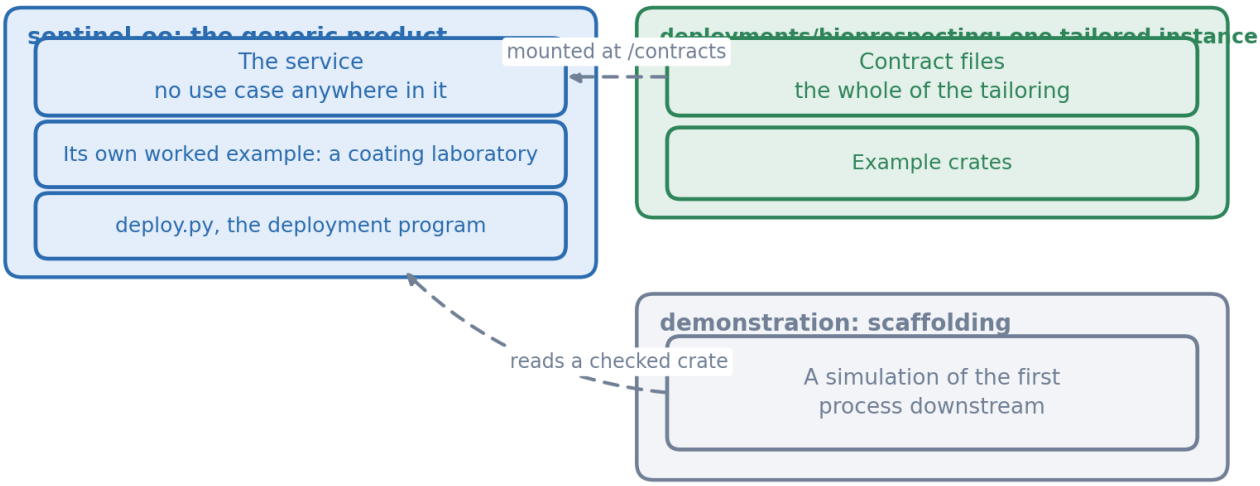


Figure 1: The three separate things

What the arrows mean, and what they deliberately do not

The product never reads anything from a deployment or from the demonstration. The dependency points one way only, and two tests fail if that ever changes.

The boundaries either side are fixed

The service is inserted into a workflow that already exists. The tool upstream and the tool downstream are real, they are not ours, and changing them is not this system's job.

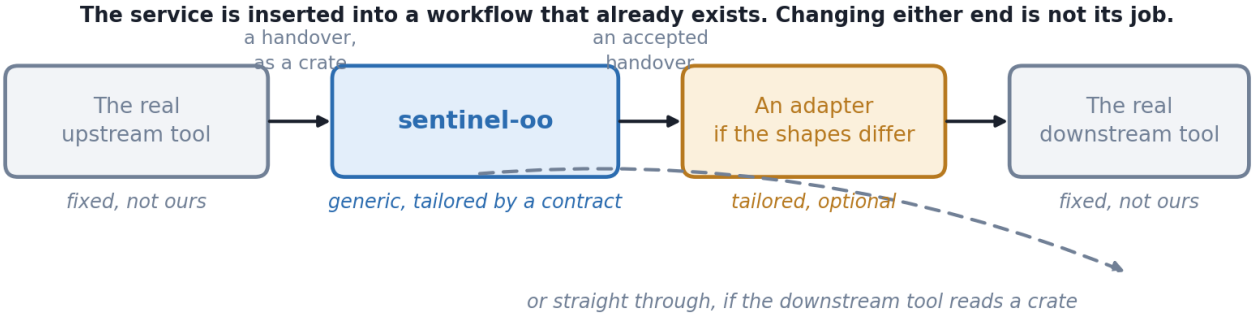


Figure 2: The boundaries either side are fixed

An **adapter** converts an accepted crate into whatever the downstream tool actually wants. It is tailored to one cycle, so it belongs to a deployment. The current demonstration has no adapter, because the simulator reads the crate directly, and that is a simplification: a real gene synthesis vendor's ordering system would not read an RO-Crate.

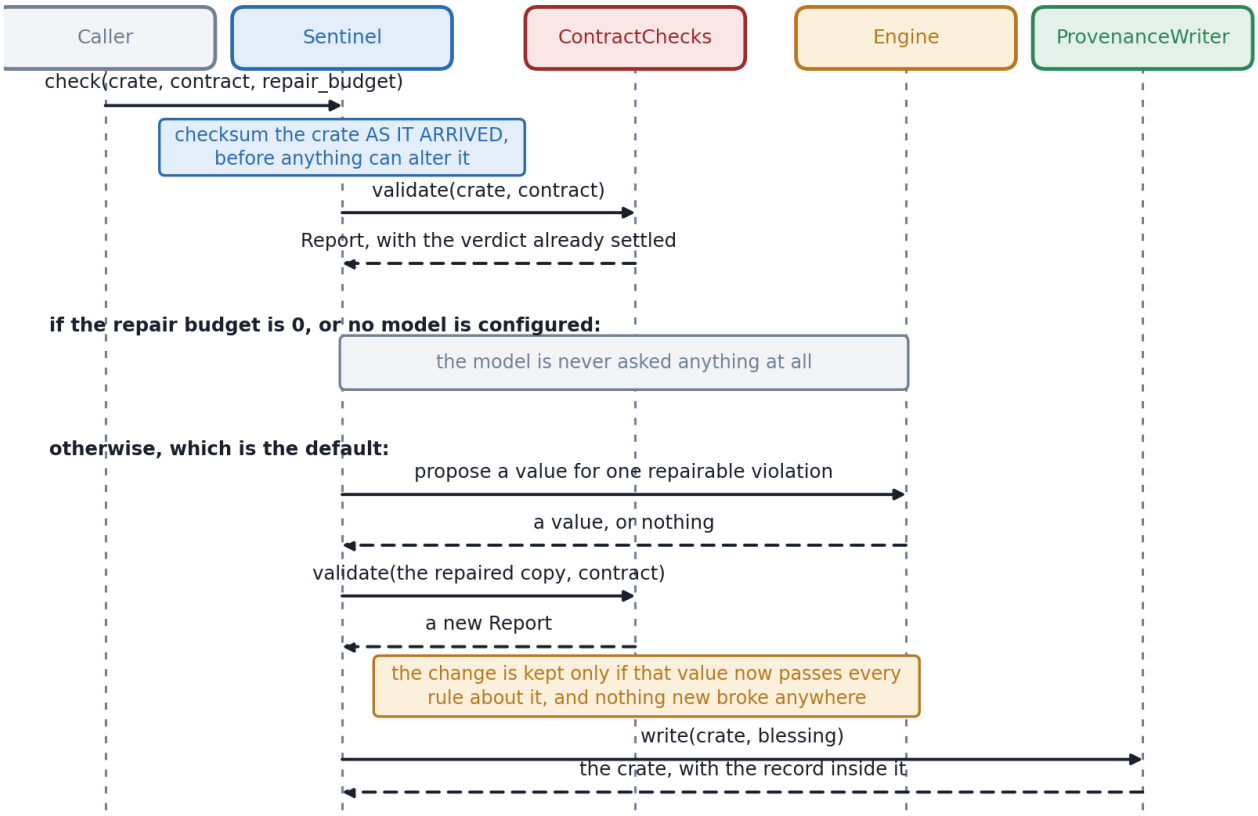


Figure 3: How one check flows

3. How one check flows

Three things about that order

The checksum is taken of the crate as it arrived, not at the end. Taken later it would describe something the sender never sent, and a reader could no longer tell whether what was blessed has since been altered.

The verdict is settled when the report is constructed and there is no way to set it afterwards. `Sentinel` never looks at the findings to form a view of its own; it carries the verdict across unchanged.

A refusal still returns a crate. A refusal that exists only as an error message in somebody's terminal is not a record. The refused crate is the artefact you want when asked in a year why a batch never went to synthesis.

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4. The components

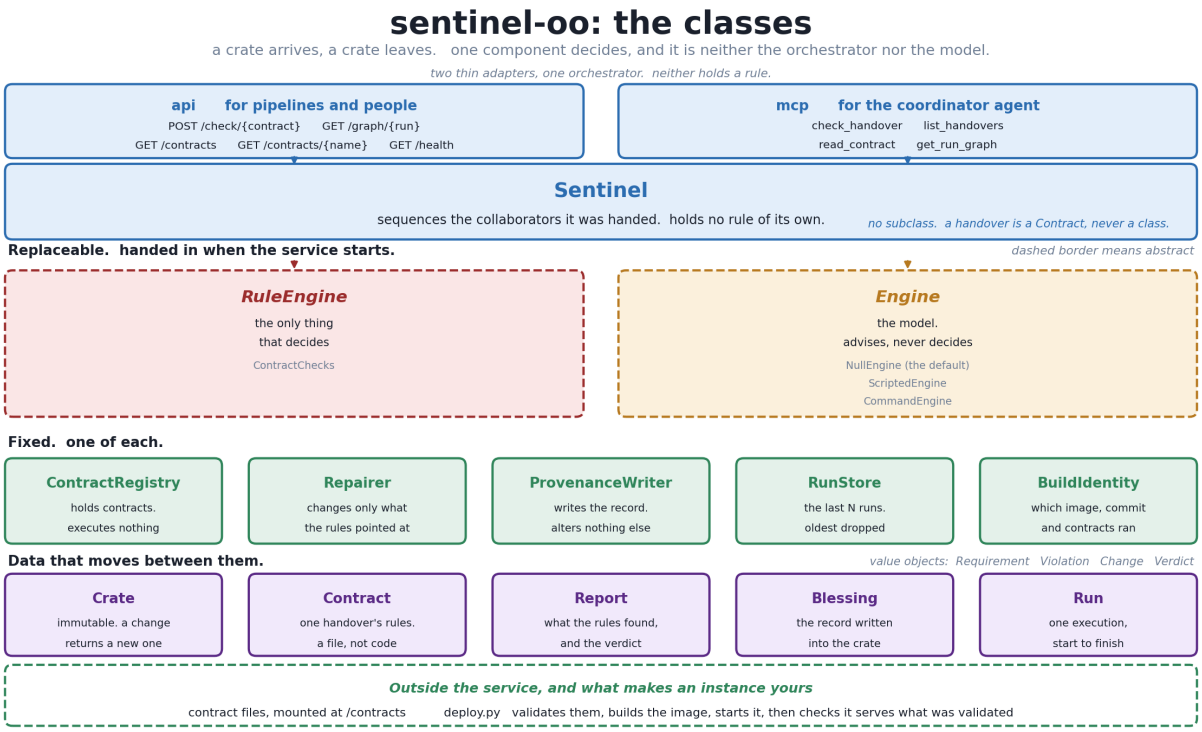


Figure 4: The classes of sentinel-oo

What each one is for

Component	Responsibility	May decide a verdict?
`Sentinel`	Orchestration. Calls the others in order and assembles the record	**No**
`Crate`	Reading, holding and writing an RO-Crate, immutably	No
`ContractRegistry`	Finding a contract by name and version	No
`Contract` and `Requirement`	What a handover must satisfy, read from a file	No
`ContractChecks`	Applying the requirements	**Yes, and only this**

Component	Responsibility	May decide a verdict?
`Report`	The findings and the verdict drawn from them	Yes, on construction
`Engine`	Asking a model one question. Advisory only	No
`Repairer`	Writing a proposed value and handing it back to the rules	No
`ProvenanceWriter`	Writing the record into the crate	No

The crate is immutable

Every operation that would change a crate returns a new one. Internally the contents are frozen: dictionaries become read-only views and lists become tuples, recursively.

That is not a style choice. The service receives data from somebody else, judges it, and returns it. If it could edit what it was given in place, then the thing it checksummed and the thing it returned would not provably be related.

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5. Where a decision may be made

This is the invariant the whole design is arranged around.

A test proves it by installing a rule engine that refuses everything and another that accepts everything, and checking that the verdict follows the engine in both cases regardless of the crate. If any crate is ever accepted while the refusing engine is installed, something other than the rule engine is deciding.

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6. A contract is data, not code

Adding a handover means writing a file. There is no class for any handover, no registration call, and no table mapping a name to code.

The vocabulary is deliberately small

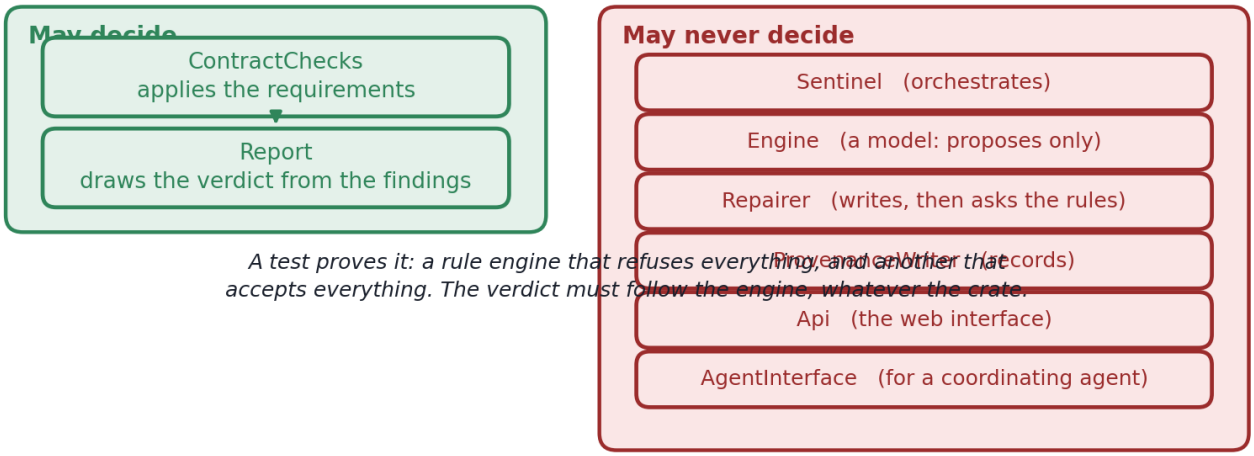
A requirement may demand exactly four things: `be recorded`, `be unique`, `point at something in the crate`, `be a number between`. A contract may also declare, with `expects`, which entity types it is written for.

The list is short on purpose. A language that can say anything can also say things nobody can check, and this list is what a contract author can be shown as the whole of their vocabulary.

Nothing is silently ignored

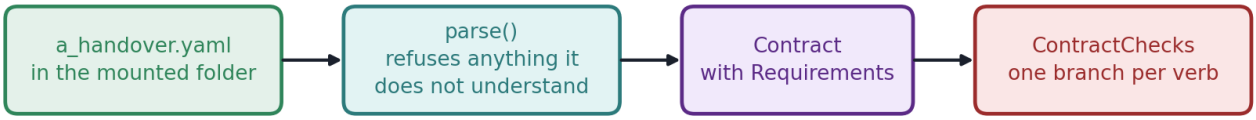
The reader refuses a contract containing a field it does not recognise, a verb it cannot check, a severity that is not one of the two, or a requirement with no message. A silently ignored requirement is a requirement somebody believes is protecting them.

In particular there is no field for instructing a model, and the error message says so. This service accepts contracts from whoever is sending the handover, so prompt text in a contract would mean the sender writing the model's instructions.



A test proves it: a rule engine that refuses everything, and another that accepts everything. The verdict must follow the engine, whatever the crate.

Figure 5: Where a decision may be made



No class for any handover. No registration call. No table mapping a name to code.

A test greps the service source for every handover name it knows, and fails the day one appears.

Figure 6: A contract is data, not code

7. The model, and what it is not allowed to do

The one job

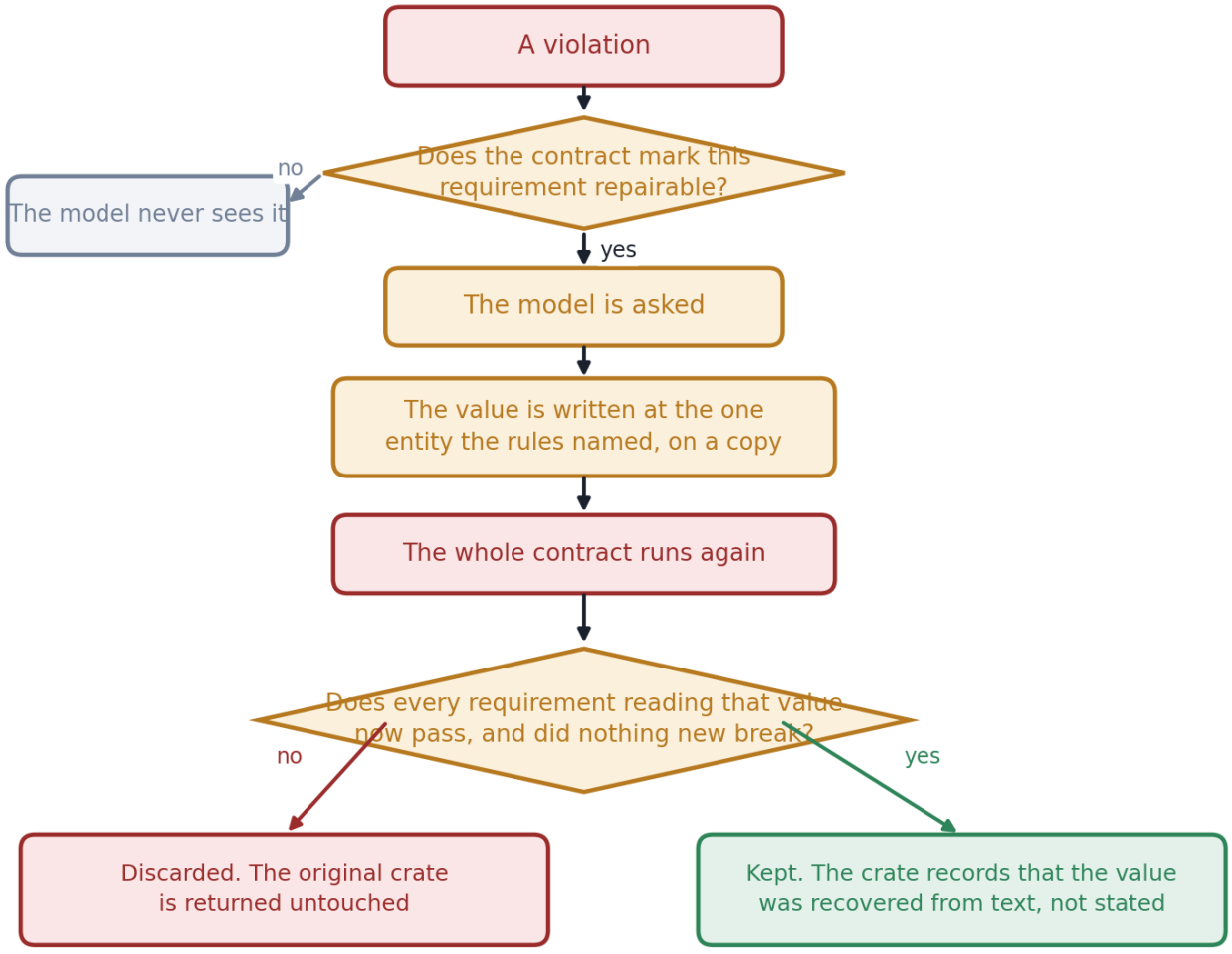
A rule can only read a value recorded as a value. If the same fact is present in the crate but only inside a sentence of free text, no rule can see it, and the handover is refused for a value that is physically there.

So the model is handed a check that has **already failed**, at a place the rules named, and asked one question: is this value already written down somewhere in this record, including inside prose?

Answering “no” is a correct and expected answer.

Where this thinking comes from. Most of this section is not new. The `sentinel` framework, which Marco Anteghini wrote for Project 25 and which sits beside this one in the same repository, settled what a model is for at a handover, and that part of it is reproduced here almost unchanged. What this design changes is how the service is structured, not what the model is allowed to do. The specific debts are listed under Acknowledgements.

What constrains it, all in code rather than in the prompt



A model is called by default, and runs in its own container, which need not be on this machine.

Figure 7: The repair path

Three further constraints:

A model is called by default, and runs in its own container, which does not have to be on this machine. A model worth using wants a GPU, and the machine with the GPU is usually not the machine running the service. Where it is and which one it is are both environment variables, `SENTINEL_MODEL_URL` and `SENTINEL_MODEL`, so a deployment points at its own without anything being rebuilt.

How long a check will wait for it, which is how long a pipeline waits. Two numbers, both settable by a deployment. The first answer is allowed 180 seconds, because the model has to be loaded before it can answer anything: measured on an NVIDIA card on 11 November 2026, that loading was about 19 seconds of a 39.6 second first answer, and slower again on a processor. Every answer after that is allowed 60 seconds, because the same question that took 39.6 seconds cold came back in 67 milliseconds warm. Only an answer shortens the wait: a call that fails proves nothing about whether the model is loaded, so the one after it is still treated as a first call.

Until 11 November 2026 both were one number, 300 seconds. A check is what a pipeline waits for, so that was five minutes a handover could block for while a model nobody could see failed to answer.

A model that does not answer in time is a fifth state in the record, kept apart from a model that was never reached and from one that looked and found nothing. A check that gave up waiting knows less than either: whether the value was there is not known. The same is true of a server that answers with something that is not this interface, such as a proxy's error page, which is read as finding nothing and recorded with the reason.

Setting the address to nothing, or passing a repair budget of 0, turns it off. That is a deliberate state rather than a failure, and `GET /health` tells it apart from a model that was wanted and could not be reached. A service with no model configured refuses and accepts exactly as one with a model does.

A proposal is applied only at the entity the rules named. Without that, a model asked about two slots claiming one position could answer by renaming a slot. The re-check would genuinely pass, because renaming does make the name unique, and a record of something that never happened would have been written.

The prompt is one fixed template. Contract text fills labelled data slots and never becomes an instruction.

The fault this design caught

Originally the code checked only the requirement being repaired. A model answered with a phrase that was not one of the permitted values. That satisfied "the setting must be recorded" and failed "the setting must be one of these", so the handover was correctly refused, **and the invented phrase was written into the crate anyway.** A refusal carrying a fabrication is worse than a plain refusal, because the fabrication outlives the refusal.

What a real model actually did

Measured on 10 November 2026 with two models running locally through Ollama, each with 14 billion parameters, where "14b" in a model's name means 14 billion parameters.

Model	Given the real job	Asked what the flag means	Told the meaning, asked to apply it
`qwen2.5:14b`	recovered nothing	confidently wrong	answered the opposite of the truth
`qwen2.5-coder:14b`	recovered nothing	confidently wrong	answered correctly

Every outcome through the service was a refusal with nothing written. Stub models standing in for wrong answers were all discarded by the rules.

What that establishes: a model of this size does not know what a bioinformatics tool's flag means and will state a wrong meaning without hedging, and a model being wrong, absent or confidently mistaken changed no verdict and wrote nothing. **What it does not establish:** that the recovery path works with a real model, because no real model got it right while doing the real job.

8. The record written into the crate

Everything the service did is written into the returned crate as RO-Crate entities, so the crate carries its own history and needs no database beside it.

What is recorded	Why
A run identifier	So two runs over one crate can be told apart
The checksum of the crate as it arrived	So anybody can tell whether what was blessed has since been altered
The contract's name, version and checksum	So a refusal can be argued about with the exact rules in hand
Where the contract came from	"Checked against the published rules" and "checked against rules the sender supplied" are different claims and must not look alike
Which build of the service ran	The image tag, the commit, and a checksum over the contracts
What happened with the model	Which of four things: none configured; reachable and asked, with how many times and how many changes kept; reachable and not asked, because no failing requirement was repairable; or configured and NOT REACHED, with the reason
Every finding	Including flags, not only refusals
Every change a model's proposal survived	With the value before, the value after, which engine proposed it, and which requirement confirmed it

The action is recorded as failed rather than completed when the handover was refused, so a reader who looks only at the action status still gets the right answer.

What the record must never let a reader conclude wrongly

A model that was not reached must not read as a model that found nothing. Until 11 November 2026 the record named the engine and said how many changes were kept, whatever had happened. A crate refused on a repairable requirement, with no model anywhere, carried `model engine: qwen2.5:14b at http://model:11434` and `0 change(s)`. Anybody reading that later would conclude a model with 14 billion parameters looked for the value and found nothing. No model was reached and the sentence holding the value was never read.

It was found by a deployment trial on 11 November 2026, in which somebody deployed the service from the documents alone and had no model running. The command line was equally silent, and only the deployment program warned, once, at start-up, in a terminal, in no record.

The record now says which of four things happened, and the fourth is the one it could not say:

`none configured`	Nobody asked for a model
`..., reachable, asked once, 1 change(s) kept`	It was asked and something survived the rules
`..., reachable, not asked: no failing requirement was marked repairable`	A good run in which the model had nothing to do
`..., NOT REACHED: <why>. No value was recovered from free text`	A model was wanted and was not there

Reachable and asked are separate facts, and so are asked and kept. A model can answer three times and have all three answers thrown away by the rules, which is a different run from one that was never consulted.

9. The two ways in

Both are thin adapters over the same orchestrator and neither contains any checking logic.

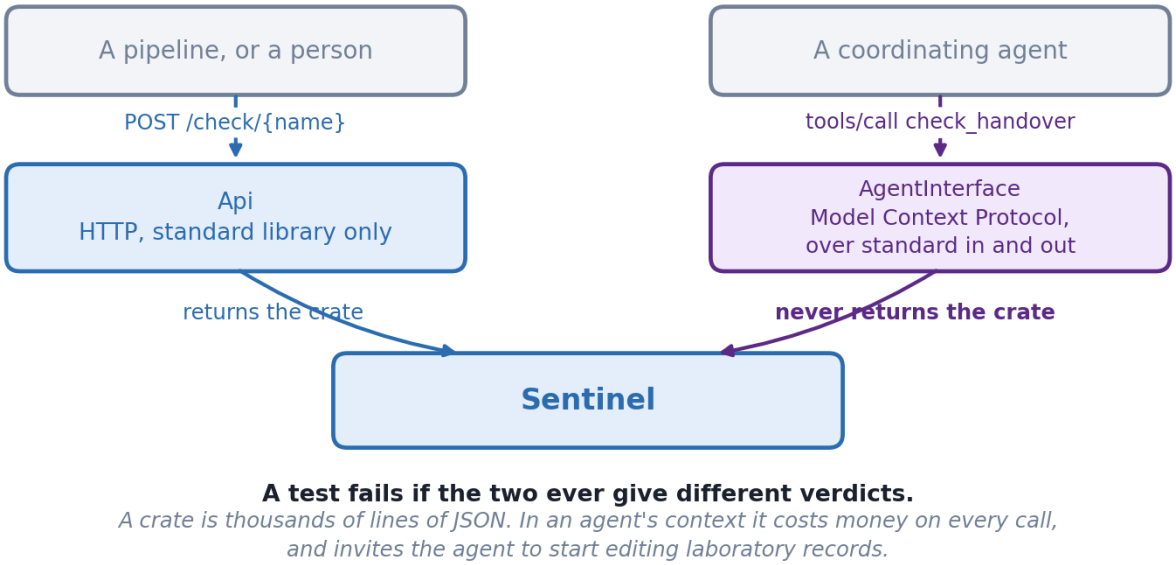


Figure 8: The two ways in

	Gets the crate back?	Why
The web interface	Yes	The crate is the thing that carries on down the pipeline
The agent interface	**No**	A crate is thousands of lines of JSON. Putting it in an agent's context costs money on every call and invites the agent to start editing laboratory records

The agent interface returns the verdict, the findings and the path the crate was written to. Its findings field is named `message_from_contract`, so an agent reading it can tell the wording came from a contract file somebody wrote and not from the service's own opinion. Its tool descriptions are fixed in code and never taken from contract files.

A test fails if the two interfaces ever give different verdicts.

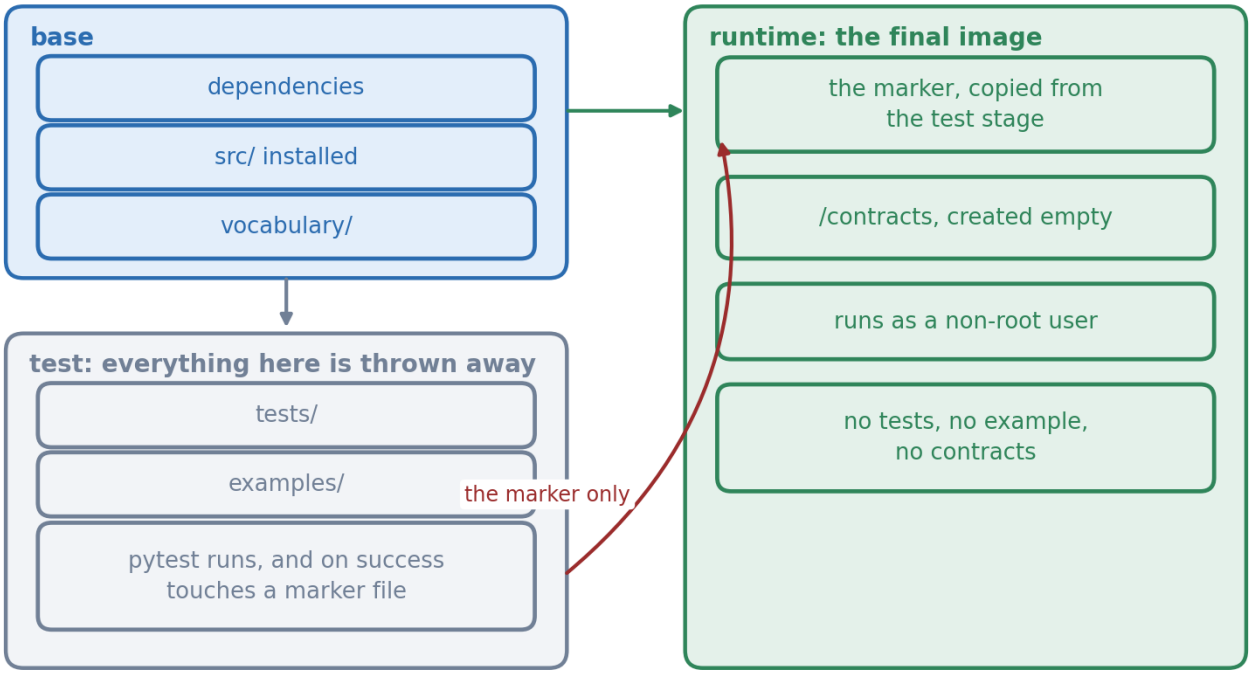
The web interface is built on Python's standard library rather than a web framework, which keeps the claim that the product runs with nothing installed true. Its route table is the single source of truth and the OpenAPI description is generated from it, so the documentation cannot drift from the service.

Full details in the API manual.

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10. The container

The marker file is the gate. It cannot exist unless the test suite passed, and the final image cannot be built without it, so an image whose tests fail is never produced. Nothing else from the test stage reaches the final image, so neither the tests nor the example ship.



The marker cannot exist unless the suite passed, and the final image cannot be built without it. So an image whose tests fail is never produced.

The build context is the product folder. Nothing outside it is read, which is what lets the product be copied out of its repository on its own and still build, test and run.

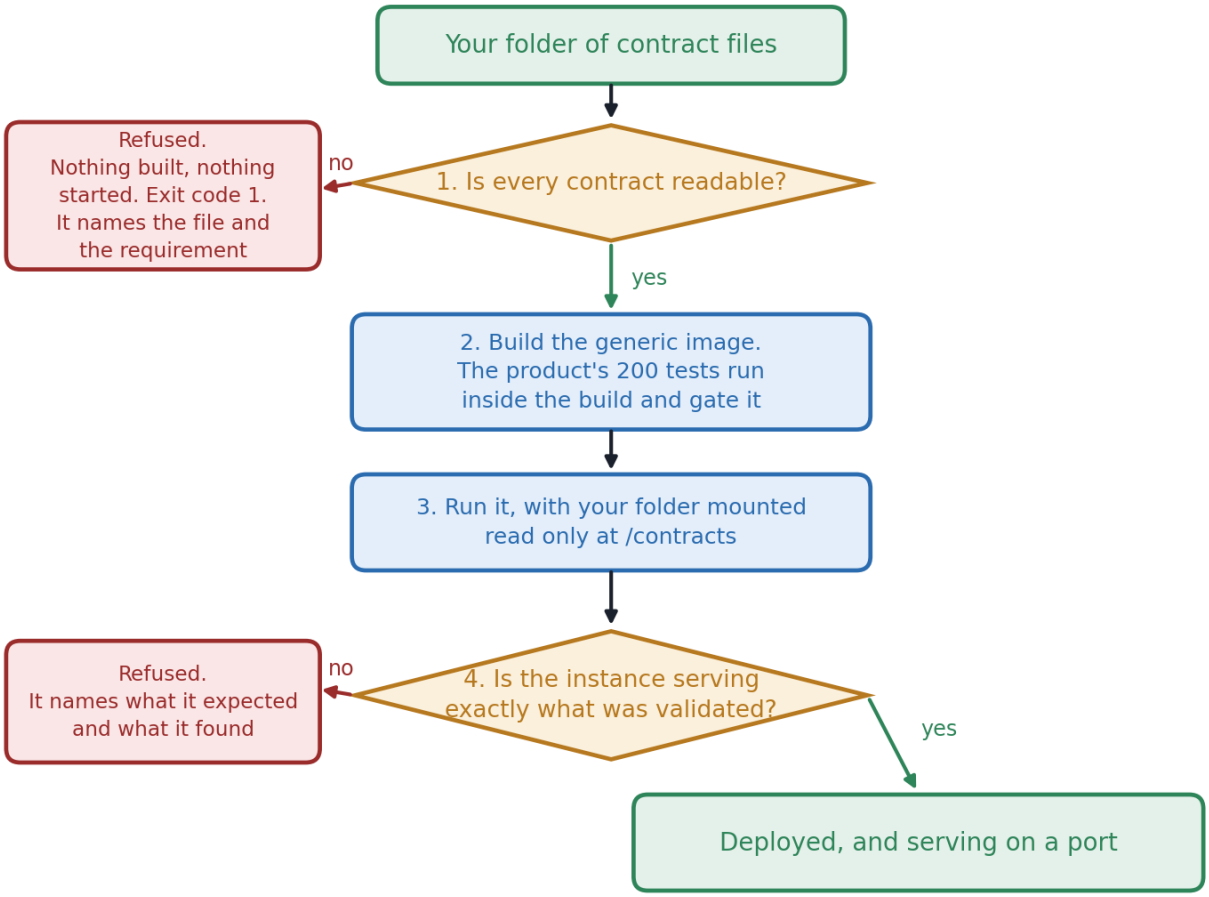
Figure 9: The three stages of the container

The build context is the product folder. Nothing outside it is read, which is what lets the product be copied out of its repository on its own and still build, test and run. That was done on 11 November 2026 by copying only that folder into an empty directory with no git repository: 200 tests passed, the image built, and the built image accepted the product's own worked example.

The final image holds no contract file anywhere, and `/contracts` is an empty mount point.

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11. A deployment



A program, not a page of instructions. No person follows steps and nothing is tailored by hand.

Figure 10: A deployment, step by step

The deployment step is a program rather than a page of instructions, so no person follows steps and nothing is tailored by hand.

Why step 1 comes before step 2. A deployment that starts with a broken contract discovers the problem at first use, which means discovering it in front of whoever sent the handover.

Why step 4 exists. It catches the wrong folder being mounted, and a contract edited between validation and start, which is a real window because the build takes about a minute. It was tested by pointing it at an instance serving a different folder, and it reported which contracts it expected and which it found.

One image, any number of deployments. Two instances were run at once on 11 November 2026, one holding bioprospecting contracts and one holding a coating laboratory's, each refusing a contract name the other one

knew. The coating contracts are now the test suite's fixtures, in `tests/fixtures/contracts/`, and they are there for exactly this reason: nothing in the service knows anything about any domain, and the same code that checks a bioprospecting handover checks a coating line with no change but the contract file.

The contracts are mounted **read only**: a service that checks handovers has no business editing the contract it checks against.

---

12. The invariants, and the tests that hold them

Each of these is a decision that would be easy to undo by accident, so each has a test that fails loudly.

Invariant	Test
Only the rule engine decides a verdict	`test_only_the_rule_engine_sets_the_verdict`
The orchestrator reaches no verdict of its own	`test_orchestrator_reaches_no_verdict`
The crate is immutable under every write	`test_crate_is_immutable_under_every_write`
The checksum is of what arrived	`test_input_checksum_is_of_what_arrived`
A refusal still returns a crate	`test_refusal_returns_a_crate`
A refusal is recorded as a failed action	`test_refusal_is_recorded_as_a_failed_action`
Every finding reaches the crate	`test_every_finding_reaches_the_crate`
The returned crate is still a valid RO-Crate	`test_the_crate_we_return_is_still_valid`, which calls a third party tool
The default path touches no network	`test_default_path_touches_no_network`
A model cannot change a verdict	`test_the_verdict_is_the_same_with_a_model_and_without`
The model is never asked about what must not be repaired	`test_the_model_is_never_asked_about_what_must_not_be_repair`
A discarded proposal leaves no trace	`test_a_discarded_proposal_leaves_no_trace`
Both interfaces give the same verdict	`test_both_interfaces_give_the_same_verdict`
The agent interface never returns the crate	`test_the_agent_interface_returns_no_crate`
No handover is named in the service source	`test_no_handover_name_in_source`
The service imports nothing from a use case	`test_source_imports_no_use_case`
No test resolves a path outside the product	`test_fixtures_stay_inside_the_product`

200 tests in the product, which run inside the image at build time. 12 more in the demonstration's simulator, which stand on their own fixtures.

Every test in the product runs with the network disabled by a fixture, so a test that quietly needs a model or a server fails rather than passing on a machine that happens to have one.

---

13. Known limitations

Recorded rather than left to be discovered.

The contract language

No comparison between two values. You cannot say that one recorded length must be about three times another. Every requirement reads one value and judges it alone. This is the first requirement met in practice that is out of reach rather than merely unwritten.

No conditional requirements. You cannot say that if one setting had a particular value then another thing must also be recorded.

No way to say "if present, must be one of these". The list of permitted values hangs off "must be recorded", so a missing value triggers both requirements and the report says two things where one would do.

Types are compared as plain text

Nothing in the service knows that schema.org considers a `Hotel` to be a kind of `LodgingBusiness`. A requirement asking for one will not accept the other. Teaching it the hierarchy means giving it the whole of schema.org.

A contract whose `applies_to` type is absent from a crate matches nothing and passes. That is correct for one requirement and was disastrous in aggregate, which is why `expects` exists. A contract with no `expects` block still has this hole, and the plain-words summary warns about it.

The service

`GET /graph/{run}` returns status 501, which in HTTP means the server does not support this. The run graph is not built. It says so rather than returning an empty graph that looks like an answer.

The agent interface has not been tested against a real Model Context Protocol client, only against the protocol as written down.

An action does not name a human or organisational agent. Process Run Crate says it should. The service is not told, so it writes nothing rather than inventing a name.

A crate with no `@context` is read without complaint. It is interpretable, so the reader accepts it. This is now expressible as a contract requirement, so it is a line somebody should write.

The product has no repository of its own, so it identifies itself by a bioprospecting repository's address. That is true today, and a wrong address would be worse than an awkward one. `SENTINEL_URL` overrides it.

With no contracts mounted, `--list-contracts` prints an empty list followed by a checksum, and that checksum is the SHA-256 of nothing at all. It should say it holds no contracts.

Not built

Advisory flags proposed by a model, and a contract authoring assistant. Both were planned as later steps. The design is arranged so that stopping where it stopped leaves a working service rather than a half-built one.

Acknowledgements

`sentinel-oo` was designed and built by Anil Wipat (Newcastle University and Prozomix Ltd) and Marco Anteghini (LifeGlimmer), for Project 25 at ELIXIR BioHackathon Europe 2026, during the week of 9 to 13 November 2026.

It also owes a great deal to the `sentinel` framework, which Marco Anteghini wrote for the same project and which sits beside this one in the same repository. This design departs from that framework in how the service is structured. It does not depart from its thinking about what a model is for, which is the strongest part of that work and is reproduced here almost unchanged.

Taken from the `sentinel` framework rather than invented here:

What	Where it appears
What a model is for at a handover, and the fact that the list is short	Section 7
That a proposal may land only where the checker pointed, and that this is enforced in code rather than requested in a prompt	Section 7, and an invariant test

What	Where it appears
The reasoning behind that rule: a model could answer a double-booked position by renaming it, the re-check would genuinely pass, and a record of something that never happened would have been written	Section 7
That an empty answer is a correct answer, and that a model which always finds something is worse than one that usually finds nothing	Section 7
The test for what must never be repaired automatically: not “is this hard to fix” but “would fixing it invent something nobody recorded”	Section 7
Vocabulary files, and the rule that a site may rename an agreed term but may not coin one	Section 10, the `vocabulary/` folder
Published example crates, including deliberately broken ones, as something to test against from the first day	Section 1, and the worked example

The structure of this service, the contract language, the deployment program, the separation of the product from a deployment, and every fault and limitation recorded in section 13 are this design’s own.